SMART INDIA HACKATHON 2025

Organization : **Government of Odisha**

PS ID : **25045**

PS Title : **Blockchain-Based Supply Chain Transparency for Agricultural Produce**

Theme : **Agriculture, FoodTech & Rural Development**

PS Category : **Software**

Team ID : **70586** | **Team Name : AgriChain001**

Team Leader Name : **Bhavithran R S**

THE PROBLEM

- Existing Supply Chains lack Transparency
- Quality and Origin of produce is not ensured
- Middlemen exploitation due to privacy in supply chain

EXPECTED SOLUTION

- A decentralized, farmer-friendly interface to trace agricultural produce
- Deployable on low-cost, trusted infrastructure
- Ensures fair pricing and reducing fraud / exploitation

HIGHLIGHTS OF OUR MODEL

- Complete transparency and authenticity over the entire Supply Chain using Blockchain
- Precise record keeping of the crop's pricing, storage and logistical data
- Reduced Middlemen exploitation due to full transparency
- Uses fast and reliable blockchain to efficiently process user transactions
- A robust marketplace for users to trade goods, reducing in-person meetings
- Simple QR scanning method for consumers to see the trace of the product

CLICK HERE TO VIEW OUR PROJECT FOLDERS

WAYMARK --- GUIDING EVERY STEP WITH TRUST

A blockchain powered platform that ensures transparent, secure, and real-time monitoring of goods across every step of the supply chain.

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Tokenizing Harvest

A popular approach to represent real world entities on blockchain

produces

Registers crop batches in app (App records to Blockchain)

mints

Blockchain mints tokens that represents crops

Click here to see sample minting on chain

Marketplace Listing

1. Growth period updates shared by farmer
2. Quality Assured with Waymark Score
3. Equipped with seamless payment gateway

delivery / takeaway of goods off chain

sell order

Listing of batches in marketplace

buys from farmer

Token transfer on chain with payment information

pays fiat through payment gateway

YouTube Solution Explanation video

Iot Powered Freight and Warehousinng

shipping goods to warehouses

SKU's IoT data logged at regular intervals

Blockchain syncs IoT data to the corresponding tokens

Tokens carry IoT data at each stage

Supply Chain View

Customers scan the QR code on the product (for the quantity they purchased) to view full supply chain

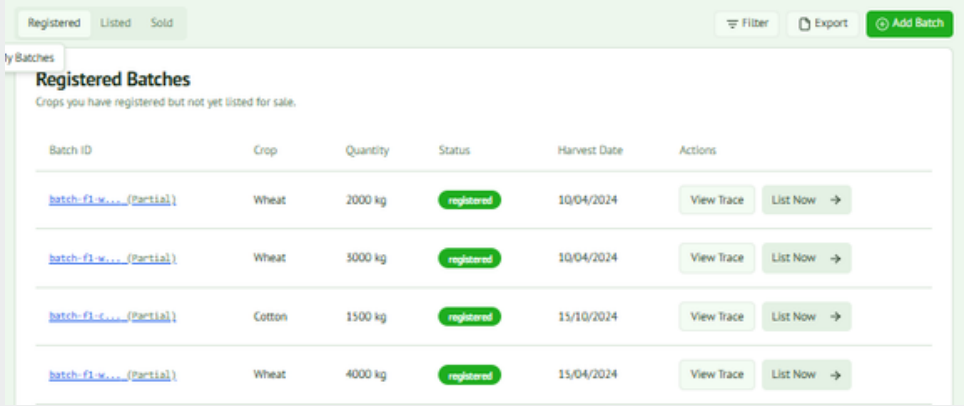
You harvested this batch — Verified
Harvested and prepared for storage. — 10/04/2024, 09:00:00 — ₹ 4,012
by Rajesh Kumar (You) — 28°C — 45% — Punjab, India

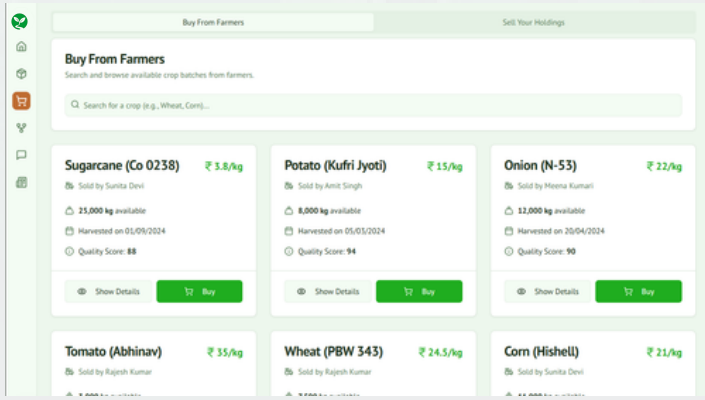
Stored at Farm — Verified
Stored in farm silo #8. — 10/04/2024, 13:00:00
by Rajesh Kumar (FARMER) — 18°C — 40% — Punjab, India Silo

Sold to Dealer — Verified
Portion of batch sold to Agro Deals Inc.. — 11/04/2024, 11:00:00 — ₹ 4,873
by Rajesh Kumar (FARMER)

In Transit to Warehouse — Verified
Transported by 'Agri-Haul'. — 11/04/2024, 13:00:00
by Agri-Haul (LOGISTICS) — 20°C — 42% — On NH 48

Delivered to Retailer Network — Verified
Delivered to distribution center in Mumbai, India. — 12/04/2024, 19:00:00 — ₹ 6,072
by Agro Deals Inc. (DEALER)





TECHNICAL APPROACH

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Blockchain Essentials

Smart Contracts
- Tokenization logic implementation
- Tracking token movement between wallets

Polygon (POL) Chain
- Faster transactions and minimal gas fee
- Ethereum-based and EVM compatible for smart contracts

Funder Wallet
- Address - 0xF8Fd742887Af3Bd4829c24fF250662D19dEdDb4C
- Sponsors gas fee for user actions throughout the application
- The seed key for all custodial wallets

Custodial Wallets
- Hierarchical Deterministic wallets for blockchain interaction
- Private-Public key pairs generated using ECC secp256k1 algorithm

Openzepplin Token Standards
- Tokenizing standard specification libraries
- Creates ERC-721 token standards (NFTs) for batch registrations
- Creates ERC-20 token standards referenced with NFTs for splitting batches

Oracle Networks
- Provides real world data to blockchain
- Helps in forecasting demand and supply of produces

IoT - Storage & Logistics

GPS Modules
- Tracks transportation of produces

Environment Monitoring
- Temparature & Humidity Sensors
- Motion & Weighing Sensors

Communication Modules
- LoRa - Longer range with low power consumption
- Cellular - 4G/5G for transporting goods across cities / countries

RFID Tags
- Tracking goods in warehouses
- Transport data (like toll charges) are transmitted

Backend & Deployment

Google Firebase
- User Authentication - Login, Timeout Session, Security

Stripe - Fiat Payment Handling
- Seamless fiat payments which triggers smart contracts through API for token transfer

Blockchain as a Service
- Storing needed data on the polygon blockchain
- Querying them back to show entire supply chain

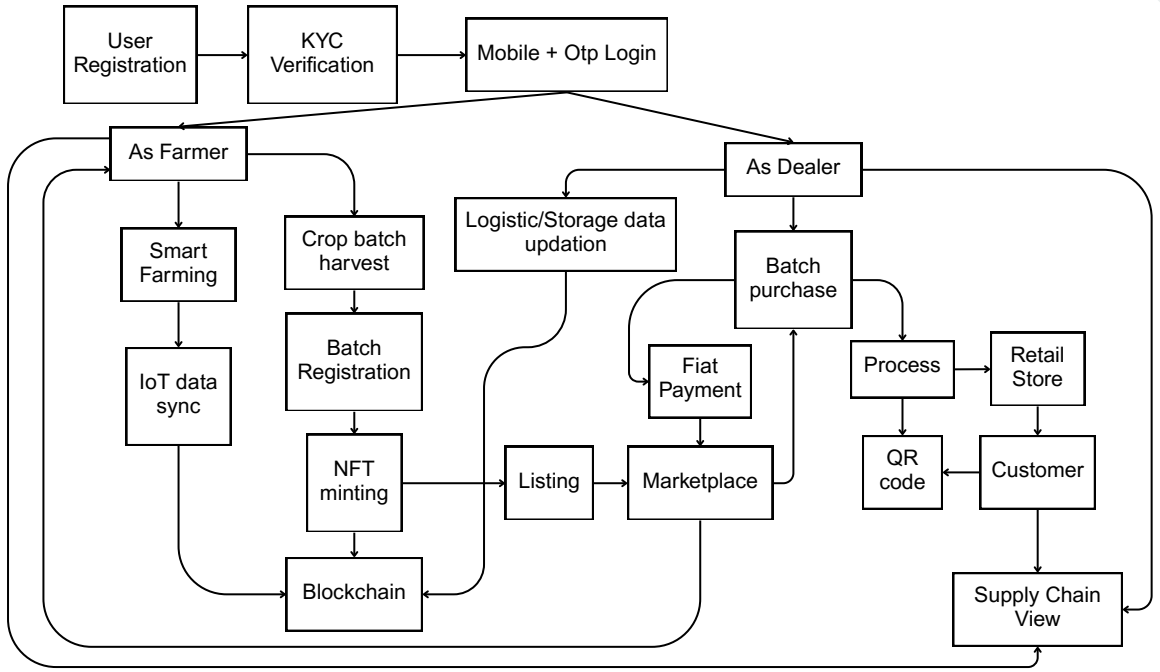
Frameworks & Tools

Frontend Languages
- HTML
- Tailwind
- Next.js
- React / shadcn ui

Blockchain Frameworks
- Web3.js
- Ethers.js

Backend Tools
- Node.js - File based data storing
- Genkit - AI powered features

Architecture Diagram



FEASIBILITY & VIABILITY

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Feasibility

Technical Feasibility

- Can handle IoT data and Blockchain with Traceability
- Local-to-Global Agri Commerce Platform
- Blockchain is already being used for supply chains

Regulatory Feasibility

- Seed and Land quality regulations
- Regional Agri laws and Government subsidies

Market Feasibility

- Transparent pricing, and fair contracts will attract farmers
- Easy farmer adoption , since easy KYC and better prices with Retail/Export support

Viability

Sustainability & Social Impact Viability

- Farmer Income: 15–20% higher realization through direct market linkages
- Cold chain & demand forecasting reduce post-harvest losses

Value Chain Viability

- For Farmers: Better prices, faster payments, affordable inputs.
- For Retailers/Dealers : Quality assurance, traceability, reliable sourcing.
- For Consumers: Transparency & trust in food sourcing.

Economic Viability

- CAC is low due to strong farmer networks and beneficial for farmers

MVP Analysis

Improvement

Enhancements

- Voice assistance in local language.
- Farmer-friendly application with minimal UI for easy usage.

Future Improvements

- Cryptographic techniques for validation of certifications.
- Automatic alerts when IoT devices malfunctions.

Github link : Click here to view Repo

MVP Report : Click here to view report

Demo video : Click here to view video

IMPACT & BENEFITS

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Fair pay and smart organic practices will increase

For Farmers

- Build reputation by building **Waymark Score.**
- Get better prices to sell.
- Can maximize yield by using planned cultivation according to demand forecasting.
- Their contribution is recognized in supply chain.
- Let's them access markets and transperent supply is created.

Prevents stockouts with little to lower operational costs

For Retailers

- Verified sourcing improves **Brand Value and Customer Loyalty.**
- Better traceability with provided QR codes.
- Readily SKU movement and price updates.

Prevents Stockpiling and adulteration

For GOVT.

- Data of entire **supply and demand projection.**
- Opportunities for exports
- **Financial transparency** on all transactions.

Optimized Inventory Management with credible Farmers

For Dealers

- Transparency brings credibility and **long term partnerships.**
- Easier to **trace issues** (e.g., spoilage, damage, contamination) at the exact stage.
- **Improves Brand Value.**

Trustful products with transparent fair price

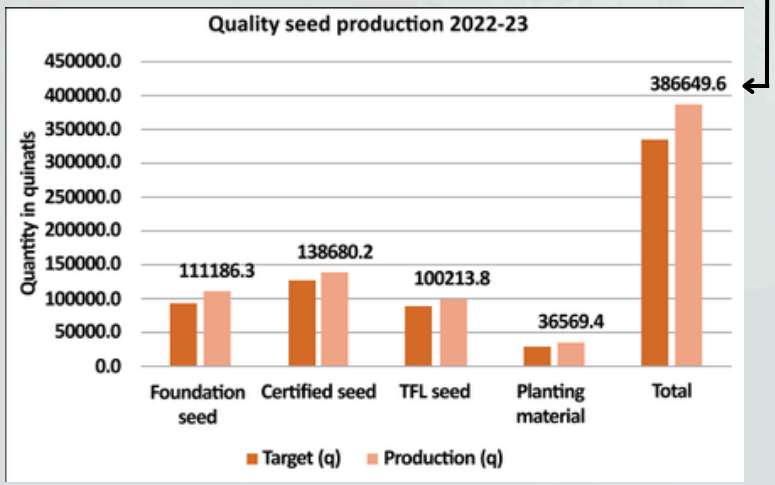
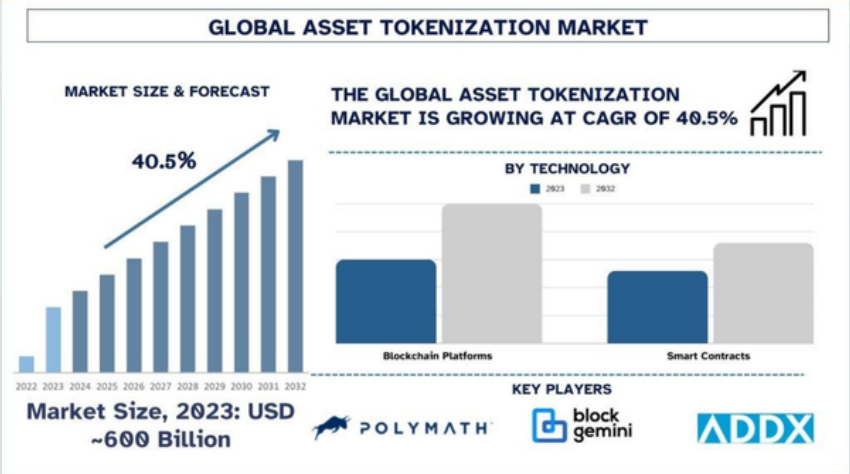
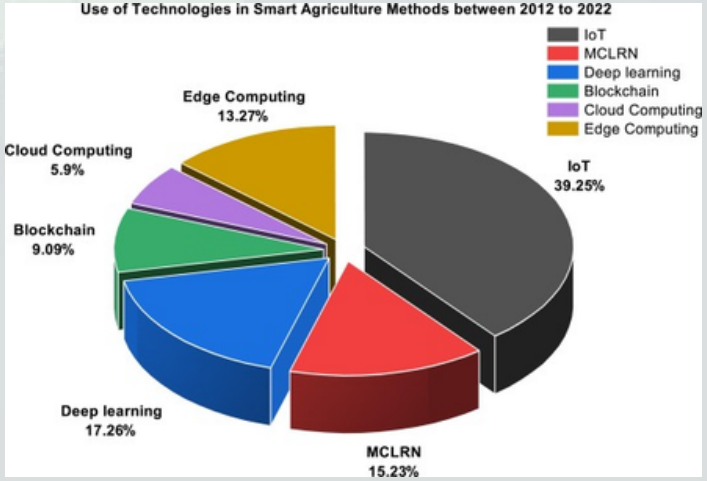
For Consumers

- Dealers/Retailers **can't mislabel adulterated products.**
- Consumers can know what they consume, it's origin and quality.
- Fair prices due to transparency and reduced middleman.
- Have confidence and awareness on what they purchase.

RESEARCH & REFERENCES

- Agriculture employs ~40% of India's workforce (World Bank, World Development Indicators, 2023).
- India has 146 million farmers (Government of India, Agricultural Census 2021).
- Post-harvest losses in India are valued at ₹92,000 crore annually (~$12 billion) due to poor logistics and storage (ICAR – Indian Council of Agricultural Research, 2022).

ICAR Report 2023 - Agricultural Research PDF

- Through our POC Survey, we came to know that most farmers can't overcome the issue of middlemen domination due to lack of awareness.
- They aren't aware of how to set price for their batches according to the quality and demand
- Introducing our platform helps them to overcome pricing issues and quality compromises through the supply chain

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- https://www.researchgate.net/publication/364580709_Blockchain_in_Supply_Chain_Management#:~:text=The%20results%20of%20the%20research,stakeholder%20of%20the%20supply%20chain.